

Q1 : What is Artificial Intelligence?

A1: Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence. AI can learn, analyze information, solve problems, understand language, and make decisions. Examples of AI include ChatGPT, voice assistants, recommendation systems, and self-driving technology.

Q2 : Difference between AI, Machine Learning, Generative AI, and Agentic AI

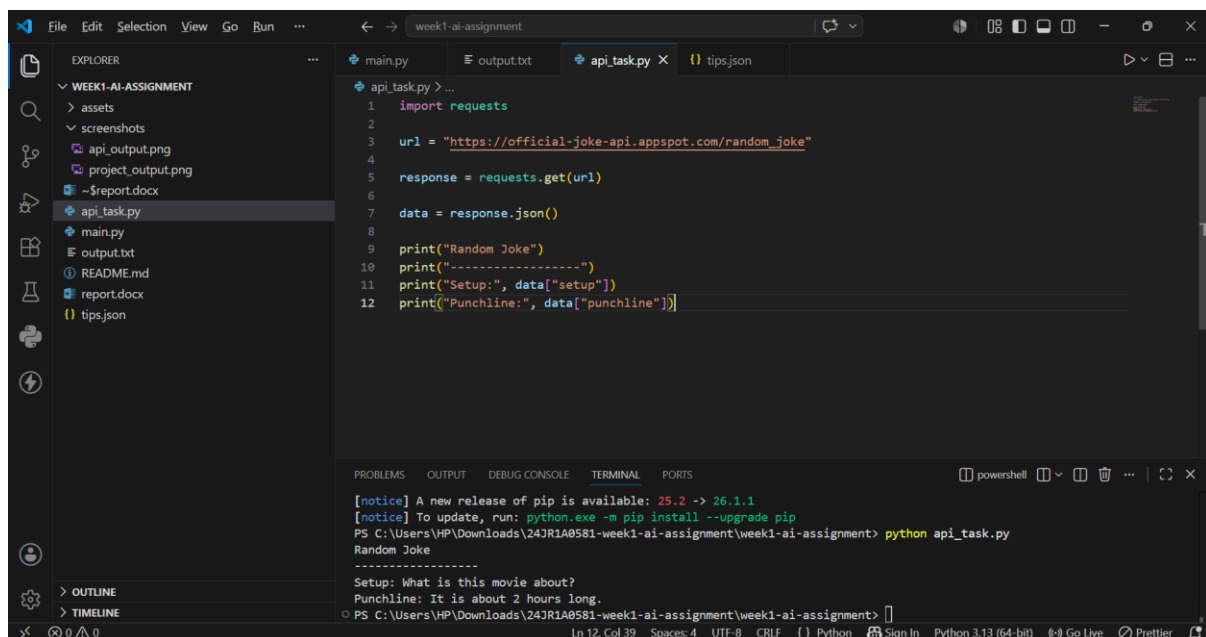
A2 : AI (Artificial Intelligence):

AI is the broad concept of making machines intelligent and capable of solving problems like humans.

Machine Learning (ML): Machine Learning is a subset of AI where machines learn from data and improve performance without being explicitly programmed.

Generative AI: Generative AI is a type of AI that creates new content such as text, images, videos, or code. Example: ChatGPT generates text responses.

Agentic AI: Agentic AI refers to systems that can independently make decisions, plan actions, and complete tasks with minimal human involvement.



The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'week1-ai-assignment' with files like 'api_task.py', 'main.py', 'output.txt', 'README.md', 'report.docx', and 'tips.json'. The code editor shows the 'api_task.py' file with the following code:

```
1 import requests
2
3 url = "https://official-joke-api.appspot.com/random_joke"
4
5 response = requests.get(url)
6
7 data = response.json()
8
9 print("Random Joke")
10 print("-----")
11 print("Setup:", data["setup"])
12 print("Punchline:", data["punchline"])
```

The terminal at the bottom shows the output of running the script:

```
[notice] A new release of pip is available: 25.2 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\HP\Downloads\243R1A0581-week1-ai-assignment\week1-ai-assignment> python api_task.py
Random Joke
-----
Setup: What is this movie about?
Punchline: It is about 2 hours long.
PS C:\Users\HP\Downloads\243R1A0581-week1-ai-assignment\week1-ai-assignment>
```

Q3 : Explain any three open-source AI models available on Hugging Face

A3 : 1. BERT: BERT is a language model used for understanding text and improving tasks like question answering and text classification.

2. Stable Diffusion: Stable Diffusion is an AI model used to generate images from text descriptions.

3. GPT-2: GPT-2 is a text generation model that can generate paragraphs, stories, and answers from prompts.

Q4 : Difference between traditional programming and AI-assisted development

A4 : Traditional programming follows fixed rules written by developers. The computer only performs tasks according to programmed instructions. AI-assisted development uses AI tools to help developers write code, suggest solutions, debug errors, and automate repetitive tasks. It makes software development faster and more efficient.

Q5 : Explain AI hallucinations with one example

A5 : AI hallucination happens when an AI system generates incorrect or false information but presents it confidently as true.

Ex: If someone asks an AI about a book that does not exist, the AI may invent the book title, author, and details even though they are fake.